

React js

History of react?

By the early 2010s, web applications were becoming increasingly complex. Traditional methods of manipulating the DOM (Document Object Model) using libraries like jQuery were proving to be inefficient and hard to maintain. Developers faced challenges in managing state, handling updates, and ensuring performance as applications grew.

At Facebook, engineers were dealing with these challenges while maintaining the News Feed, a highly dynamic and interactive feature. They needed a solution that could simplify UI development and improve performance.

In 2011, Jordan Walke, a software engineer at Facebook, developed the initial prototype of React. He was inspired by concepts from XHP, a PHP library for building user interfaces, and thought about applying these ideas to JavaScript.

The prototype, originally called "FaxJS," was used internally at Facebook to build parts of the News Feed. The success of the prototype encouraged further development, and React was officially released to the public at JSConf US in May 2013.

Why do we need react js ?

React js is used to make our work more easier as compared to java Script.

Because with the use of react js we can make our work done in less code

1. React js is java Script library that is used to build the fast and interactive web /mobile applications.
It provide robust and optimize way to build an modern application.
2. It work on JSX based i.e It is a combination of html and java Script.
3. It is an open source ,i.e we can see react js content present in it ,like its file,component ,codes etc .
4. React js is reuseable component based frontend library for java Script.
5. There are two types of component in react function and class component.
React suggest to use function component to do your work because it contain daily-updated or on-goeing update by react. But,we have to learn both because company can have a old project that are created in class components .

Note:

what do you mean by reuseable component based?

➤ Ex:- to understand component:

In a web application can be its header or ,fotter or , its image ,or it can be anything on web application.

- For ex to understand why it is reuseable :
- if you have made header of website ,and you want to use same header for all other web pages you can call it from react js.

That's why it is called java Script library.

6. React uses virtual dom concept to efficiently update the actual dom.

Instead of directly updateing the dom for every changes ,react create the virtual dom ,first representation of dom in memory and calculate the most efficient way to update the actual dom.

Note: as we know dom refers to doucument object model which consist of html components.

- **Virtual DOM:** The Virtual DOM was one of React's most innovative features. It acts as a lightweight copy of the real DOM, allowing React to determine the minimal set of changes required to update the UI i.e user_inteface . This approach significantly improved performance.
- **React uses virtual dom to optimize rendereing.**
- Render refer to showing code on web pages .

How to install react js?

Step1:

Search node.js on bowser >

Click get node.js@ >

Click Window installer>

To check the version of node install you can do:

Open Cmd (command prompt) > type: node -v

To check version of node packages manager (npm: manages pacages present in node_modules):

Open cmd > type: npm -v

Step2:

Create folder > click on path of folder & clear it any type cmd> then ,
Type: `npx create-react-app my-app` > then, enter .>

Then, it will show you the message at last after lodeing “ happy hackeing”,

Note: if error comes after above code done then,

Type:

`npm install -g npm@latest`

>then, type this : `npx create-react-app my-app`

Now , finally to check recat is completely installed you can go to

➤ Folder created > click on my-app > you can see files

How to start react in vs code ?

Folder>my-app>click path >cmd>type: “ code . ” >

Click on terminal >new terminal >type: `npm start`

Note: terminal is beside the view at top header of vs code inside (...)

Explain the each file present in my-app folder ?

README.md:

Consist of react cammand desicription . like `npm start` etc..

Package.json:

Consist of whole project information like:
dependencies, name ,version ,script etc..

3.node_modules:

It consist of file regardeing the dependencies which are in package.json

Q> if the node_modules is not there or deleted ?

> command to create it is using package.json because as we now package.json consist of all the react information is:
npm install;

4.package-lock json:

it consist of all information regardeing the dependencies which are in package.json

5.gitignore:

the things you do not want to upload on GitHub you can keep in a gitignore. I will ignore that part of project to push it on gitHUB .

note: gitHUB - it is platform you can upload your side of project & also get others project sides crested by other peoples by using password /user id create my company or form you are workeing for.

7. src:

it contains mutiple file like:-

reportWebVitals.js : it determine performance of project

index.js: reportWebVitals.js it is call inside it.

Index.css : it is used to style the index.js

App.js : it is link to index.js and index.js link to index.html in public file.

7.public:

it contains mutiple file like:-

manifest.json: it is used to create progressive web-applications i.e application which without internet show previous data.

How to create custom command of inplace of “npm start” or to change any other?

First go to package.json >

You can see object reacted named “script”

>inside this object you can see that “start” is key given

> now, you can change this key named “start” to whatever you want

For an ex: if you have changed the named to “ronack”

npm ronack; // this will give error

to run a custom command will :

npm run ronack;

difference between npm and npx?

Npm stands for :node pakacege manager, it is used to manages node packages prasant node_modules.

Npx : node package execution , it used to execute node packages

Present in node_modules. And to create the project. Ex: command

npx create react-app my-app; // my-app is the name given to project

what do mean by the state in react js ?

In React, the **state** refers to an object that holds information about a component's current situation. This information can change over time, typically as a result of user actions or **data fetching**, and when **state changes**, React **re-renders** the **component** to reflect the **updated UI with the help of using useState()**.

Whenever state changes, React re-renders the component to reflect the updated data. This enables you to build dynamic UIs that respond to user interactions.

What is the use of curly{} brackets in react?

In React, **curly braces {}** are used in JSX to embed JavaScript expressions within HTML-like syntax. This allows you to dynamically include variables, functions, or objects in your components.

What do you mean by the props?

In React, **props** (short for "properties") are used to pass information from **one component to another**. The main purpose of props is to allow a **parent component** to send data to its **child components**. And type of prop is object.

What do you mean by The preventDefault() method?

The preventDefault() method is a crucial part of event handling in JavaScript. It is used to **cancel the default action** that belongs to an event, provided the event is cancelable. This method is commonly used in scenarios where you want to prevent the default behavior of an element, such as submitting a form or following a link.

What do mean by react js life cycle method?? Vvv.imp

It consist of three phases in their working order:

phase 1: mounting

ex:- suppose loadeing or opeining any website on browser.

Mounting consist of three methods in their working order:

- `constructor();`

it is always driven from parent class using `super()` which is React.

- `render();`

it is used to reflet / show API data from `componentDidMount();`

- `componentDidMount();`

it is used to get API data .

we put API data in this method, because it work after other two methods.s



.

NOTE: While React itself is not asynchronous, it works seamlessly with JavaScript's asynchronous features, such as Promises, `async/await`, and APIs like Fetch. This is particularly useful when fetching data from servers or handling real-time updates.

Synchronous programming, also known as blocking or sequential programming, executes the tasks in a predetermined order, where each operation waits for the previous one to complete before proceeding.

Asynchronous programming allows tasks to execute independently of one another, enabling concurrent execution and improved performance.

phase 2: updateing

ex;- like addeing any product to cart (add to cart) in amazon website.

NOTE: THE DATA added to cart is stored in local storage. So, the data remains in the cart after closeing the website.

➤ It consist of methods like:

➤ **comoponentDidUpdate:**

it run after the state or component is updated.

You can castamize it with providing condition to it

It consist of three parameter to get previous props , previous state and snapShot, in same order .

Snapshot depends upon method `getSnapshotBeforeUpdate()` which has its own parameter previous props , previous state and it return the same things .

➤ **shouldComponentUpdate():**

it return a Boolean value ,accordeing to which the state is update .

phase 3: unmounteing :

it consisit of method like `componentWillUnmount()` that will work before dom is removed

ex: _

means removeing data from dom / local storage after the order is placed.

Q life cycle method in function component???

RACT JS LIFE CYCLE IN FUNCTION COMPONENT :--

`useEffect` is a hook imported from react library ,

it takes two parameter call back function & parameter

it will work after the dom been rendered.,just like componentDidMount() in class.

it is used to directly update data in dom.

//consist of same phases :-

1.mounteing: it is done using useEffect() method

2.unmounnteing: it is done inside return in the useEffect method.

3. updateing: also, it is done using useEffect() method

Q> diff b/w useLayoutEffect() && useEffect().

use layout effect= work after the DOM has been updated but

before the browser paints the screen i.e when useEffect work.

use effect= work after the DOM has been updated

Q>what is the use of the pure component?

Pure component are used in class components

In React, a **PureComponent** is a type of component that optimizes performance by reducing the number of render operations that need to occur. It achieves this by implementing a shallow comparison of the component's props and state. If there are no changes detected in the props or state, the component does not re-render.

Q> use Memo?

USE MEMO= (it is just like we use PureComponent in class component)

it is hook that is used to stop uncessray re-execution while re-rendering .

useMemo is used to memoies value to avoid unassessary re - calculation.

usememo updates when its dependencies are updated

it is used in function component.

Q> what do mean by ref ?

Refs are used to access and modify the **DOM elements** in the React Application. It creates a reference to the elements and uses it to modify them.

In class component to use ref we use “createRef()” function

& In function component to use ref we use “useRef()” function

Refs are a function provided by [React](#) to access the [DOM element](#) and the React elements created in [components](#). They are used in cases where we want to change the value of a child component, without making use of [props](#) and [state](#).

Q> What do you understand by MEMO? And Difference b/W use memo & MEMO ?

using Memo will help you create a component which will not re-render when its parent re-render as long as prop is not updated.

1.use memo:

- it is used in function component.
- it is a hook
- useMemo is used to memoize value to avoid unnecessary re-calculation.

2.MEMO:-

- it is used in both function and class components.
- it is high order function
- it using memo will cause react to skip re-rendering

of component if its props has not change.

What& When we use useCallback ?

note: memoization is a optimization technique that is used to prevent unnecessary re-rendering of components

.

use call back :- it is use to return a memoize callBack function

it is use to avoid memoise funtion to

prevent unnecessary re-creation. return memoize function

when we pass proops with functionn we use usecallBack method.

beacuse it have to parameter call back function & a dependency as array .

diff b/w useMemp & useCallback??

useMemo: is used to memorize 'value' to prevent unnecessary re-calculation

useCallback: is used to memoize 'function' to prevent unnecessary re-creation

what do understand by routeing??

routing:

for diffrent-diffrent functionality we need diffrent-diffrent

web pages wich we run on unique/diffrent url which we called routeing

- To use routing in react you have add its dependency to package .json
Using command > npm i react-router-dom.

>then,

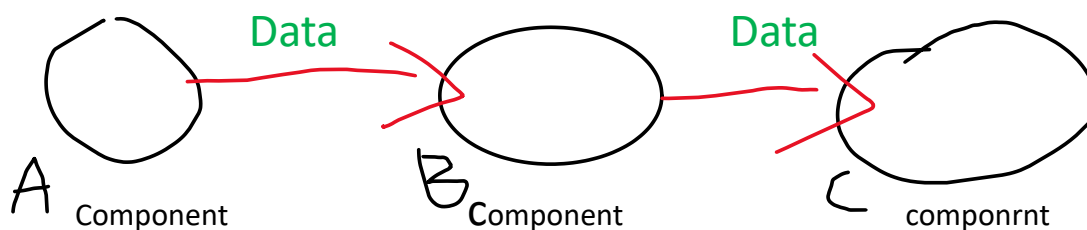
```
import {BrowserRouter, Routes, Route, Link} from "react-router-dom";
```

explain the concept of props drilling?

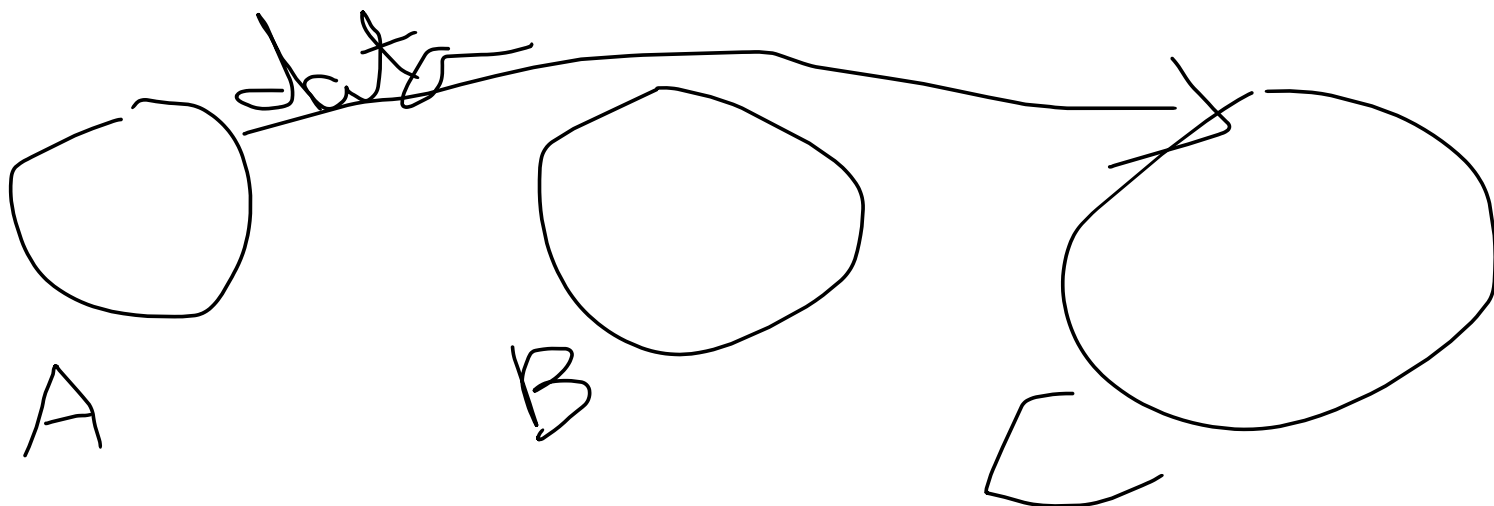
It refer to process of passing data /or functions through multiple level oof nested components by using props, even when the intermediate component doesnot using it directly use those props.

1.with prop drilling :

For ex;- a to c data/function transfer .b will get unnecessary data in between.



2. to solve this essue we use contextAPI >



// CONTEXT API :

// it includes method like >>

//1.createContext() : use to create context

//2. useContext(): use to get context created.

// we can pass data to multiple component using context api,

//it is an alternative way of props drilling.

Q> StudentUseReducer (hook) ??

it is similar to useState hook.

it is used make custom logic .

instead of using multiple state we can use StudentUseReducer.

ex:in the below program:--

in useReducer it consist of two parameter reduce and initialState

where reduce is function which consist of two parameter state and action

in which action communicates with reduce and reduce communicates with dispatch

.i.e action->reduce->dispatch

and IN State it communicates with initialState and initialState communicates with data .

i.e , state->initialState -> data.

HOW TO MAKE NEW PROJECT FROM START??

Before below steps install node js .

1.Create folder

2. click folder url & remove it, and type cmd

3.in cmd type : **npx create-react-app my-app**

4.after installeing , go inside folder > myapp >remove url>type cmd> type code.

5.src folder >App.js file> remove content from <header> to </header> inside div.

6. type the follweing in terminal to install its dependencies in package.json folder
> npm i react-router-dom

Used for routeing

> npm i axios

To fetch apis

> npm i json-server

7. create db.json file in my-app > copy the fake any api data which is in the form of array of objecf just like below and and paste it on db.json file you have created.

```
{
  "students": [
    {
      "id": "2003",
      "studentName": "DHRUV _SINGhh",
      "studentEmail": "ronack@gmail.com"
    },
    {
      "id": "2004",
      "studentName": "nitishs",
```

```
"studentEmail": "nit@gmail.com"
},
{
  "id": "332c",
  "studentName": "new name",
  "studentEmail": "newemail@gmail.com"
}
}
```

8.now run db.json using command:

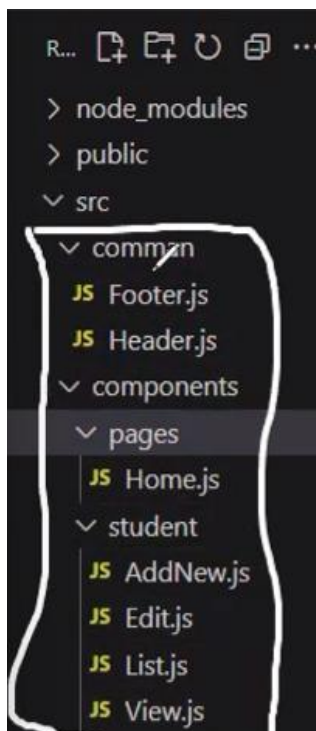
`npx json-server db.json` or `npx json-server db.json --port 3333`>

Copy the Endpoints:

<http://localhost:3000/students>

and,paste it on any browser to get this fake api data .

8.Now create the folder and file just like shown in img below-



Now, you can make CURD application using this..

CRUD stands for **Create, Read/Retrieve, Update, and Delete**
Every application is made using this.

Topic 2: REDUX

Q> what do you mean by redux?

It is state management tool.

It is a npm package use for comunication b/w components just like context api ,but we have to install it.

But,in case of context api it is in-built which we import from react library.

Life cycle of redux;

View -> action-> root reducer -> store

- View : it consist of html element
- Action: like buton on clicl etc..
- Root reducer: it is a root of multiple reducer with different methods ex;add to cart
- Store: it refer to storage like local and session storage.
- Note: local storage is used to store data perment unless you manually delete it,
- But,Session storage is used to temporary data

Now, create this folder and file same in your vs code as shown in image :-



Topic 3: TypeScript

Q>what do mean by typescript?

TypeScript is the new JavaScript. Yeah, You have heard right. TypeScript, which is the superset of JavaScript has achieved a surge and is becoming the go-to language for the development of every large application. The feature of detecting the bug while typing code makes it unique. As per the research, almost **60% of JavaScript developers already use TypeScript.**



Providing syntax for types is a key feature of TypeScript, with which its name has been implemented "**TypeScript**". Even tech giants like Microsoft and Accenture are migrating toward TypeScript because of its functionalities and features. Communicating tools like Slack and Asana use TypeScript for their implementation. Developers who work on technologies like Angular, React, and Vue use TypeScript. For better productivity, TypeScript is preferred because it is better for collaboration. So, when there is so much about TypeScript, you should know the reason why you should pick TypeScript over JavaScript.

How to use it on vs code?

1.Go to folder path type :

On cmd type : `npm install -g typescript,` then, code .

2.inside vs code :

Create file , file_name.ts

Run this command on terminal : npm install -g ts-node

You can run typescript directly using : npx tsx file_name.ts

To check the version of typescript : tsc -v

How to run it on browser:

1.compile: tsc file-name.ts

2.node file-name.js

